

PathAble

AI-Powered Smart Campus Navigation for Differently Abled Students

Product Feature Brief | Hackathon Edition | 2026

20	4	15	3
Total Features	Priority Tiers	App Screens	AI Integrations

An AI-powered, fully software-based campus navigation web app that gives differently abled students personalized, accessible, real-time route guidance, obstacle alerts, and emergency support — all from a single smartphone browser with zero hardware required.

The Problem

Differently abled students on college campuses face daily navigation challenges: inaccessible routes, broken elevators, no obstacle warnings, and no fast emergency help. Google Maps is not accessibility-aware and doesn't know campus interiors.

The Solution

PathAble delivers a personalized, AI-guided campus navigation system adapting to each student's disability type. Fully software-based, runs in any browser, zero hardware needed.

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PRIORITY 1 — Must Demo to Judges

These 6 features form the core demo loop. Every judge interaction should flow through them. They must work flawlessly on demo day.

F01

Disability Profile Setup and Personalization

- Multi-step onboarding: select disability type (mobility/wheelchair, visual, hearing, cognitive, combo)
- Mobility aid selection: manual wheelchair, electric wheelchair, crutches, white cane, none
- Navigation output mode: voice only, haptic only, visual only, or all combined
- Profile stored in localStorage — no backend or medical database needed
- All personalization logic runs client-side
- Profile editable any time in under 60 seconds from settings

F02

Accessible Route Navigation with Campus Map

- Interactive campus map with Leaflet.js and custom GeoJSON overlay of campus paths
- Every path tagged: surface type, slope (flat/mild/steep), width, step count
- Wheelchair routing removes stairs and steep slope edges before calculating path
- Dijkstra's algorithm implemented in JavaScript, runs entirely client-side
- Route shown as highlighted polyline with step-by-step turn instructions
- Three route options: shortest distance, fewest obstacles, most rest stops

F03

AI Navigation Assistant (Claude API)

- Chat-style assistant powered by Anthropic Claude API on the navigation screen
- Voice input via browser Web Speech API — student speaks naturally
- System prompt contains full campus knowledge base: buildings, rooms, facilities, obstacle reports
- Claude returns plain-language answers displayed in chat and read aloud via SpeechSynthesis
- Auto-generates accessible step-by-step checklist when crisis or obstacle is reported
- Full pipeline: browser mic → Web Speech → Claude API → SpeechSynthesis — zero hardware

F04

Real-Time Obstacle Reporting and Crowdsourced Alerts

- Any user reports: Elevator Down, Wet Floor, Blocked Ramp, Construction, Missing Signage, Other
- Location selected from campus zone dropdown — no GPS hardware required
- All active reports appear as red warning pins on the shared Leaflet map instantly
- Reports expire automatically after 4 hours unless confirmed by another user
- If 3 users confirm same report, routing engine recalculates avoiding that zone
- Admin can dismiss or escalate any report from the admin dashboard

F05

One-Tap SOS Emergency Alert

- Large red SOS button permanently visible on the bottom navigation bar
- 5-second countdown confirmation modal prevents accidental triggers
- On confirm: push notification to all admins with student zone and disability type
- Live incident timer starts on admin dashboard immediately
- Nearest first aid location and safe zone highlighted on student's map after trigger
- SOS log stored in database for admin welfare follow-up

F06

Admin Crisis Command Dashboard

- Separate /admin dashboard — admin-role access only
- Live three-column layout: obstacle reports, campus map, SOS alerts
- Each incident card: reporter, zone, issue type, elapsed time, AI severity badge, action buttons
- Admins push campus-wide alerts appearing as red banners on all student screens within 30s
- Key metrics: active incidents today, SOS events this week, most-reported zones
- Monthly report download generates structured CSV of all incidents

PRIORITY 2 — Build and Show in Demo

These features extend the core and demonstrate depth. Build them after P1 is solid. Each one should be demonstrable in the live demo flow.

F07

Crowd Density Heatmap

- Heatmap overlay on campus map showing predicted crowd density by zone
- Density pre-calculated from hardcoded class schedule timetable JSON
- Current hour from browser clock applies matching density: green / amber / red zones
- Crowd-sensitive profiles (autism, anxiety) see auto-banner suggesting quieter routes
- Fully time-table-driven — no live sensors, updates every hour automatically

F08

Sensory Safe Room Finder

- Dedicated screen listing all quiet/sensory rooms on campus with availability
- Each entry: building, room number, floor, equipment, opening hours, live status badge
- Occupancy tracked via voluntary digital check-in — student taps "I'm Here"
- Status badge: Available (green), Partially Occupied (amber), Full (red)
- "Navigate Here" button launches accessible route from student's current zone
- One-tap shortcut on home screen — no searching required

F09

Timetable-Aware Route Notifications

- Student enters weekly class schedule: subject, building, room, day, time
- 15-minute advance browser push notification: "Time to leave for [Subject]"
- Buffer adjusted by disability: wheelchair users get 20 minutes, others 15
- Tapping notification opens app to pre-calculated accessible route for that class
- Schedule stored in localStorage — no backend integration needed
- Auto-email to faculty when navigation to class starts (if enabled)

F10

Safe Zone Checkpoints on Routes

- Pre-seeded rest spots, benches, covered areas, medical rooms as GeoJSON points
- Blue shield icons on Leaflet map always visible
- Route panel lists every safe zone checkpoint along path with name and distance
- "Rest-Friendly Route" toggle switches algorithm to prioritise checkpoint-heavy paths
- Students can submit new rest spots — admin approves before going live

F11

Gamified Reporting System

- Points awarded: report submitted (5 pts), confirmed by 3 users (15 pts), confirming another (3 pts)
- Profile completion earns 10 pts, 5-day usage streak earns 20 pts
- Badges: First Responder, Safety Guardian, Community Pillar, Navigator, Campus Ready
- Monthly leaderboard of top 10 contributors with anonymised display names
- Badge collection shown as visual grid of earned and locked badges on profile
- Badge unlock triggers confetti burst via canvas-confetti library

F12

First Aid and Emergency Resource Map

- All first aid kits, medical rooms, and AED locations pre-seeded as map points
- Red cross icons always visible at all zoom levels on the Leaflet map
- Clicking any icon: resource name, stock list, last inspection date, Navigate Here button
- Admin can update resource status: Available / Needs Restock / Out of Service
- When SOS triggered, three nearest emergency resources pulse-animate on student's map

PRIORITY 3 — Unique Differentiators That Win Judges

These features are the ones competitors won't have. They demonstrate genuine product thinking, technical ambition, and deep empathy for the target users.

F13

High Contrast and Colour-Blind UI Modes

- 6 themes: Standard, High Contrast, Deuteranopia, Protanopia, Tritanopia, Full Monochrome
- Each theme is a pre-built CSS class applied to body — instant switch, no reload
- All map icons use shape + label differentiation, not colour alone
- Theme preference saved to localStorage and applied on every app load
- Theme preview strip in settings shows each theme before selection

F14

Faculty Accommodation Notification

- Student optionally enters faculty email per course during timetable setup
- When navigation to class starts, automated email sent via EmailJS free tier
- Email: "A differently-abled student is navigating to your [Room] class at [Time]. Please ensure accessible seating is prepared."
- Fully opt-in per course — toggle in profile settings enables or disables
- Log of all sent accommodation notifications shown in student profile

F15

Offline Mode with Cached Map Data

- Service worker caches: campus GeoJSON, all static routes, last obstacle reports, safe rooms, emergency resources
- Offline banner appears: "You are offline — showing last synced map data"
- Navigation, route calculation, and map rendering continue fully offline
- SOS offline mode shows pre-saved emergency contacts as static display
- On reconnect, queued obstacle reports auto-sync to Supabase

F16

AI Incident Severity Classifier

- Every obstacle report and SOS description sent to Claude API with classification prompt
- Claude returns JSON: severity (critical/serious/minor), category, recommended_action
- Severity badge stored with report and shown on both admin dashboard and student alerts
- Recommended action shown to admin as suggested first-response step
- Makes admin dashboard genuinely AI-powered, not just a CRUD interface

F17

Live Campus Announcement Banner

- Admin posts campus-wide announcement: title, message, type, expiry time
- Appears as dismissible banner on all student screens within 30 seconds via polling
- Colour-coded: red (safety), amber (maintenance), blue (general information)
- Read aloud once via SpeechSynthesis for students with voice mode enabled
- Students tap to expand and see full message with timestamp

F18

Student Wellbeing Check-In

- Daily check-in prompt: "How are you feeling today?" — Energetic / Moderate / Low Energy / Stressed
- Low Energy automatically activates rest-friendly routing for the day
- Stressed/Anxious activates quiet route mode avoiding busy zones, shows nearest sensory room
- Check-in under 5 seconds — date-stamped so it never repeats the same day
- Weekly mood chart on student profile using Chart.js

F19

Barrier Reporting with AI Auto-Description

- After selecting issue type and location, student can tap "AI Describe" instead of typing
- Claude API auto-generates clear, professional description of the likely accessibility issue
- Example: "Elevator in Block C East Wing appears non-functional. This creates a complete access barrier for wheelchair users on floors 2-4."
- Student can edit the auto-generated description or use it as-is
- Ensures consistently worded, actionable reports in admin dashboard
- Removes typing burden for students with cognitive or motor impairments

F20

Accessible Campus Induction Tour

- 5-step guided tour with animated tooltip overlays on first login
- Tour covers: Profile Setup → Map Navigation → SOS Button → Report an Issue → Safe Rooms
- Each step narrated via SpeechSynthesis for visually impaired users
- Student can skip, pause, or restart from the Help section
- Completing tour awards 10 bonus points and the "Campus Ready" badge

Tech Stack & Architecture

Layer	Technology	Why
Frontend	React 18 + Vite	Fast build, component-based, ideal for responsive SPA
Styling	Tailwind CSS	Utility-first, fast polished UI, no custom CSS files
Maps	Leaflet.js + React-Leaflet	Free, open-source, no API key required
Map Tiles	OpenStreetMap	Free tiles, no key: {s}.tile.openstreetmap.org
Routing	Dijkstra's (custom JS)	Client-side on campus GeoJSON graph, no library
AI	Anthropic Claude API	Voice assistant, incident classifier, auto-description
Voice Input	Web Speech API	Built into Chrome/Edge — no library needed
Voice Output	SpeechSynthesis API	Built into all modern browsers
Charts	Chart.js	Wellbeing mood chart and admin metrics
Email	EmailJS (free)	Faculty accommodation alerts, no backend email server
Offline	Vite PWA + Workbox	Service worker and asset caching
Database	Supabase (free tier)	PostgreSQL + JS client, real-time, all app data
Auth	Supabase Auth	Email/password, free tier, student and admin roles
Notifications	Browser Notification API	Push alerts and timetable reminders, no third party
State	React Context API	No Redux — keeps project simple and fast to build
Deployment	Replit hosting	Built-in, zero config required

Database Schema Summary

Table	Key Columns
users	id, email, full_name, role (student admin), created_at
profiles	id, user_id, disability_types[], mobility_aid, nav_modes[], emergency_contacts, faculty_notify_enabled, theme, points, badges[]
obstacle_reports	id, reporter_id, zone, issue_type, description, ai_severity, ai_category, ai_recommended_action, confirmation_count, status, expires_at
sos_alerts	id, student_id, student_name, disability_types[], zone, status (active resolved), resolved_at
announcements	id, admin_id, title, message, type (safety maintenance general), expires_at
safe_rooms	id, name, building, floor, equipment[], opening_hours, status, occupancy_count, approved
emergency_resources	id, name, resource_type (first_aid aed medical_room), building, floor, zone, status, last_updated
wellbeing_checkins	id, user_id, mood (energetic moderate low_energy stressed), date
accommodation_alerts	id, student_id, subject_name, room, faculty_email, sent_at

Row Level Security (RLS): Enabled on all tables. Authenticated users can read all reports and announcements. Users can only insert/update their own profile and reports. Only admin-role users can update resource/safe room status and post announcements.

UI & Design Specification

Color Palette

Hex	Name	Usage
#2563EB	Primary Blue	Buttons, active states, route lines
#10B981	Emerald Green	Success, available badges, safe zones
#EF4444	Danger Red	SOS button, critical alerts, emergency
#F59E0B	Warning Amber	Serious alerts, heatmap medium, obstacle pins
#F8FAFC	Background	Main app background
#0F172A	Text Primary	Near black, all primary text
#64748B	Text Secondary	Slate grey, body and helper text

Layout & Typography

- Font: Inter from Google Fonts — 400 body, 600 headings/labels, 700 app name and SOS only
- Student app: mobile-first, max width 430px centred on desktop with phone-frame shadow
- Admin dashboard: full-width desktop layout — sidebar or top nav, three-column content grid
- Base padding: 16px on all screen edges. Card gap: 12px. Section gap: 16px
- Cards: rounded-2xl (16px radius), shadow-sm, 1px border in #E2E8F0

Key UI Components

Component	Specification
Bottom Nav Bar	Fixed, 5 icons with labels. Active = primary blue filled. Inactive = slate grey
SOS Button	Rightmost nav item, bright red, white text, continuous CSS pulsing ring animation
Toast Notifications	Top-right, auto-dismiss 4s, colour-coded by type
Loading States	Skeleton shimmer loaders on map and dashboard
AI Chat Panel	Slide-up from bottom, CSS transition, handle bar to drag-close
Modals	Centred overlay, semi-transparent backdrop, rounded-2xl, max 400px wide
Map Route Line	#2563EB, 5px stroke, animated dashed flow suggesting movement direction
Obstacle Pins	Red circle with exclamation icon

Safe Zone Pins	Blue shield icon
Emergency Pins	Red cross icon always visible at all zoom levels